



Ensemble of feature selection algorithms: a multi-criteria decision-making approach

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Abstract

For the first time, the ensemble feature selection is modeled as a Multi-Criteria Decision-Making (MCDM) process in this paper. For this purpose, we used the VIKOR method as a famous MCDM algorithm to rank the features based on the evaluation of several feature selection methods as different decision-making criteria. Our proposed method, EFS-MCDM, first obtains a decision matrix using the ranks of every feature according to various rankers. The VIKOR approach is then used to assign a score to each feature based on the decision matrix. Finally, a rank vector for the features generates as an output in which the user can select a desired number of features. We have compared our approach with some ensemble feature selection methods using feature ranking strategy and basic feature selection algorithms to illustrate the proposed method's optimality and efficiency. The results show that our approach in terms of accuracy, F-score, and algorithm run-time is superior to other similar methods and performs in a short time, and it is more efficient than the other methods.

Keywords Multi-criteria decision-making · VIKOR method · Ensemble feature selection · Feature ranking

1 Introduction

With the advancement of new technologies such as the Internet of Things (IoT) and social networks, we see vast amounts of data being generated daily and at high speed. These data are generally high dimensional, i.e., contain many training samples or many features [1]. In many cases, the data include irrelevant and redundant features. The irrelevant features do not affect achieving the output, and the redundant features are almost a mix of other features and do not provide new information. The existence of such features in datasets makes learning algorithms time consuming, and it also affects their performance. Feature selection is used as an effective way to deal with these problems. Feature selection

as a dimensionality reduction technique chooses meaningful features and removes the irrelevant and redundant ones from the dataset.

This procedure can reduce data dimensionality, computational cost, time complexity, storage complexity and increase classification accuracy [2–4]. From two perspectives, we can categorize feature selection methods: supervision and search strategy [5]. From the supervision perspective, these methods can divide into three main classes: supervised, unsupervised, and semi-supervised [6]. A supervised feature selection method selects features based on label information. Supervised methods are applying to Single-Label (SL) [7, 8] and Multi-Label (ML) data [9–14]. Unlike SL, each sample contains more than one label in ML data, and the correlation between features and labels is considered feature relevance [6, 15]. Semi-supervised methods work with a type of data that includes some labeled samples [16, 17]. Unsupervised feature selection methods are methods that work with unlabeled samples [18].

Based on the search strategy, feature selection methods also can be partitioned into three main groups: filter, wrapper, and embedded. In filter methods, the statistical measures such as Fisher Score [6], Mutual Information (MI) [19], ReliefF [20], Symmetric Uncertainty (SU) [21], and Information Interest [5] use to evaluate the features before

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the learning process [22, 23]. On the other hand, wrapper methods assess the features in the learning procedure. These methods assess possible feature subsets to determine the best values according to their predictive power [2, 24]. Finally, embedded methods seek to address the weaknesses of the two previous techniques and use the strengths of both [5].

Alongside these methods, a new technique called ensemble feature selection was recently introduced. In this method, the output of several feature selection methods aggregates to obtain the final feature subset. The rationale of this approach is based on the old proverb "Two heads are better than one" and that the variety of different methods makes the feature selection method work better than only one method [25, 26].

In the real world, many issues consist of multiple objectives requiring optimization. These multi-objective optimization issues occur in many engineering and medical problems and need to chase various goals simultaneously. Different approaches are used, such as metaheuristics [27–37], Pareto-based methods [38, 39], Multi-Criteria Decision-Making [40], to deal with optimization problems. MCDM is a powerful tool for defining the preferences and achieving desired outcomes based on multiple decision-makers [41, 42]. Chai et al. [43] summarized the developments in using decision-making techniques in supplier selection. Kim et al. [44] present a new VIKOR method that used a new weighting measure for objective weights and ranks alternatives based on this new measure. Acuña-Soto et al. [45] proposed a new approach to rank instructional videos based on videos' quality from a multidimensional perspective using the VIKOR method.

Ensemble methods can be used to improve the robustness of FS algorithms. Each FS algorithm may find feature subsets that can be considered local optima in the feature subsets space. Ensemble feature selection can help address this problem by combining multiple FS algorithms' outputs and improving accuracy. On the other hand, MCDMs are methods in which decisions are made based on several decision-makers' opinions to choose the best alternatives. In this regard, we have modeled the ensemble feature selection process into an MCDM problem in which features are considered as alternatives and FS methods as criteria.

Given that in MCDM methods, each criterion has a different effect on the outcome, each target's value shows the different results of each target. Therefore, in our way, we set equal weight for all FS methods. In our proposed approach, the VIKOR method is used as one of the most popular methods for solving MCDM problems. The VIKOR method evaluates the alternatives based on a decision matrix considering the multiple FS methods' output as our criteria. Also, existing algorithms in ensemble feature selection still do not address all the challenges in this area. Among the weaknesses of existing algorithms, we can refer to low accuracy in high-dimensional datasets, long execution time, and

tuned by specific feature selection methods. In this paper, we have tried to address these issues with an adaptive and fast algorithm. This technique is the first work that models the ensemble FS problem to an MCDM process to the best of our knowledge.

An overview of our proposed method is as follows:

- The user can determine the number of selected features.
- This algorithm is an ensemble of filter methods.
- The VIKOR method is used for finding the most relevant features.

To illustrate the proposed method's effectiveness and efficiency, we have compared our approach with some similar techniques and base FS algorithms on different real-world datasets. The results indicate that our proposed method yields better results than the other methods in terms of accuracy, F-score, and algorithm run-time.

This paper's structure is organized as follows: In Sect. 2, we reviewed some related methods in ensemble FS and MCDM. In Sect. 3, we introduce the basic concepts of our proposed algorithm, and then in Sect. 4, we discuss the proposed algorithm in detail. The experimental results have been presented in Sect. 5. Finally, in Sect. 6, we concluded our work and outlined our future path.

2 Related works

In this section, we will review some recent works in Ensemble FS and MCDM. So far, several methods have been proposed for ensemble FS. These methods use different strategies such as fuzzy sets [46, 47], simple aggregation methods [48, 49], evolutionary algorithms [50, 51], multi-objective optimization [52, 53], and clustering [49]. Ansari et al. [47] proposed an ensemble FS method for sentiment classification. They used the concept of hesitant fuzzy sets to combine the results of different filter FS methods. This approach selects top-k ranked features based on the relevancy score.

In [46], a technique based on Maximum Relevancy and Minimum Redundancy (MRMR) is presented for ensemble FS using Hesitant Fuzzy Sets (HFSs). This algorithm is a filter-based method, and the results of different rankers are combined using the concept of information energy of HFSs. In [49], several ensemble FS methods are presented based on some basic techniques like max, min..., and election methods. They also offered several clustering-based methods using the mean-shift algorithm. In [54], an optimal two-dimensional ensemble feature selection algorithm is proposed to deal with feature aggregation and stability issues. In [51], a weight adjustment procedure is applied to each base feature selection algorithm's results in the ensemble process. However, existing algorithms in ensemble feature

selection still do not address all existing challenges. Among the weaknesses of existing algorithms, we can refer to low accuracy in high-dimensional datasets, long execution time, and tuned by specific feature selection methods. Therefore, this area is still one of the challenges in feature selection.

MCDM algorithms are useful approaches to handle optimization problems and issues requiring ranking. New algorithms are proposed based on the MCDM process, such as approaches in [55–57]. Zhang et al. [58] presented a group decision-making approach in which each expert's relations over alternatives are based on fuzzy logic. This model is offered feedbacks based on bounded confidence levels and the leadership of experts. MFS-MCDM [36] is a new approach proposed for reducing the dimensionality of multi-label data. In this method, the features are evaluated based on their relationship with class labels in a multi-criteria decision-making process. In [59], an approach for large-scale group decision making is proposed based on hesitant fuzzy linguistic information. In this technique, all the assessments of decision-makers are based on fuzzy linguistic term sets. After balancing them by Linguistic Distribution Assessment (LDA), a clustering approach is used to cluster decision-makers. Based on these results, linguistic metrics are combined to achieve the final assessment of each decision-maker.

3 Fundamental concepts

3.1 Ensemble feature selection

Ensemble feature selection is the process of combining multiple models instead of one model. The rationale of this approach is based on the old proverb "Two heads are better than one" and that the variety of different methods

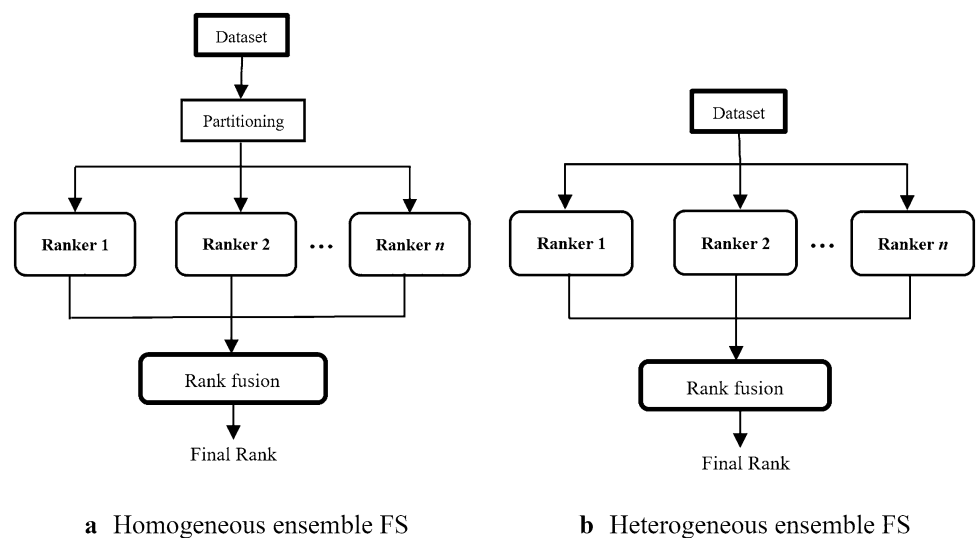
makes the feature selection method work better than only one method. Various studies have shown that these methods usually achieve better accuracy than individual methods because of the diversity of these approaches and control of variance. Ensembles have recently been used in machine learning techniques such as supervised classification, regression, and optimization. In the area of feature selection, these methods have also recently been widely used.

Ensemble FS methods can be categorized into two main groups: The first group is called homogeneous ensembles, which exploit data diversity. In these methods, the data is partitioned into multiple partitions, and a feature selection method is implemented on each division. Finally, each partition results in achieving the final feature subset (Fig. 1a). The second group is called heterogeneous ensembles, which exploit the diversity of functions. In these methods, multiple feature selection methods are executed on the data, and the results of these FS methods aggregate to find the best subset of features (Fig. 1b) [25, 60].

3.2 Multi-criteria decision making and VIKOR algorithm

Multi-Criteria Decision Making (MCDM) techniques have been popular nowadays and are useful in many real-life problems. These methods are usually used to address complex decision problems with conflicting objectives. It refers to a situation when we have a set of options in real-life, and we want to select the most preferred one. In an MCDM, there is a group of decision-makers (DM) with unique characteristics, such as experience, skills, knowledge, etc. In most cases, the DMs may have different abilities and interests. Thus, each DM should have a different degree of influence on the results. MCDM methods try to rank alternatives in subjective preference order [41, 61, 62].

Fig. 1 Block diagram of homogeneous and heterogeneous ensemble FS



One of the most popular methods to solve MCDM problems is VIKOR (Serbian: Vlsekriterijumska Optimizacija I Kompromisno Resenje) [63]. This method tries to rank alternatives closer to the ideal solution, like another MCDM method called TOPSIS [64]. These two methods seek maximum group utility and minimum regret [65].

Then the utility measure (S) and the regret measure (R) are calculated as follows:

$$R_i = \max_j \left[w_j \frac{(X_j^* - x_{ij})}{(X_j^* - X_j^-)} \right], S_i = \sum_{j=1}^T w_j \frac{(X_j^* - x_{ij})}{(X_j^* - X_j^-)}, \quad (3)$$

Algorithm 1: Template of VIKOR algorithm

Input: $M \times T$ decision-matrix X , and $1 \times T$ weights vector w

Output: VIKOR index values Q_i

1. $X^* = \max(X)$ # Obtain maximum value of each column in X
 2. $X^- = \min(X)$ # Obtain minimum value of each column in X
 3. # Calculation of utility measure (S) and regret measure (R)
 4. **for** $i = 1:M$
 5. **for** $j = 1:T$
 6. $S(i, j) = (w(j) * ((X^*(j) - X(i, j)) / (X^*(j) - X^-(j))))$
 7. **end**
 8. **end**
 9. $S_i = \text{sum}(S, 2)$ # Obtain sum of each row in X
 10. $R_i = \max(S, 2)$ # Obtain maximum value of each row in X
 11. # Calculation of VIKOR index (Q_i) for each feature
 12. $S^* = \max(S_i)$ # Obtain maximum value in S_i
 13. $S^- = \min(S_i)$ # Obtain minimum value in S_i
 14. $R^* = \max(R_i)$ # Obtain maximum value in R_i
 15. $R^- = \min(R_i)$ # Obtain minimum value in R_i
 16. $v = 0.5$ # initial the maximum group utility
 17. $Q_i = (v * (S_i - S^-) / (S^* - S^-)) + (1 - v) * ((R_i - R^*) / (R^* - R^-))$
 18. Rank the alternatives based on Q_i in ascending order
-

VIKOR method takes a weights vector $w \in \mathbb{R}^{1 \times T}$ and a decision matrix $X \in \mathbb{R}^{M \times T}$ as inputs, where M refers to the number of alternatives, T refers to the number of DMs (criteria). The best and worst values for all criterion are determined as the first step:

$$X_j^* = \{ \max(x_{ij}) | i = 1, 2, \dots, M \} = \{ X_1^*, X_2^*, \dots, X_T^* \} | j = 1, 2, \dots, T, \quad (1)$$

where x_{ij} is the value of the i -th criterion for alternative i , X_j^* refers to the maximum value of j -th criteria, M and T also represent the number of alternatives and criteria, respectively.

$$X_j^- = \{ \min(x_{ij}) | i = 1, 2, \dots, M \} = \{ X_1^-, X_2^-, \dots, X_T^- \} | j = 1, 2, \dots, T, \quad (2)$$

where X_j^- refers to the minimum value of j -th criteria.

where R_i and S_i indicate the regret and utility measures for alternative i , respectively. Besides w_j is the weight of the j -th criterion.

Finally, the VIKOR index (Q_i) for each alternative is obtained by the following equations:

$$S^* = \max_i S_i, S^- = \min_i S_i \quad (4)$$

$$R^* = \max_i R_i, R^- = \min_i R_i. \quad (5)$$

$$Q_i = \frac{v * (S_i - S^-)}{S^* - S^-} + \frac{(1 - v) * (R_i - R^*)}{(R^* - R^-)}, \quad (6)$$

where v and $1 - v$ denote the weight for maximum group utility and the weight of the individual regret. When $v > 0.5$,

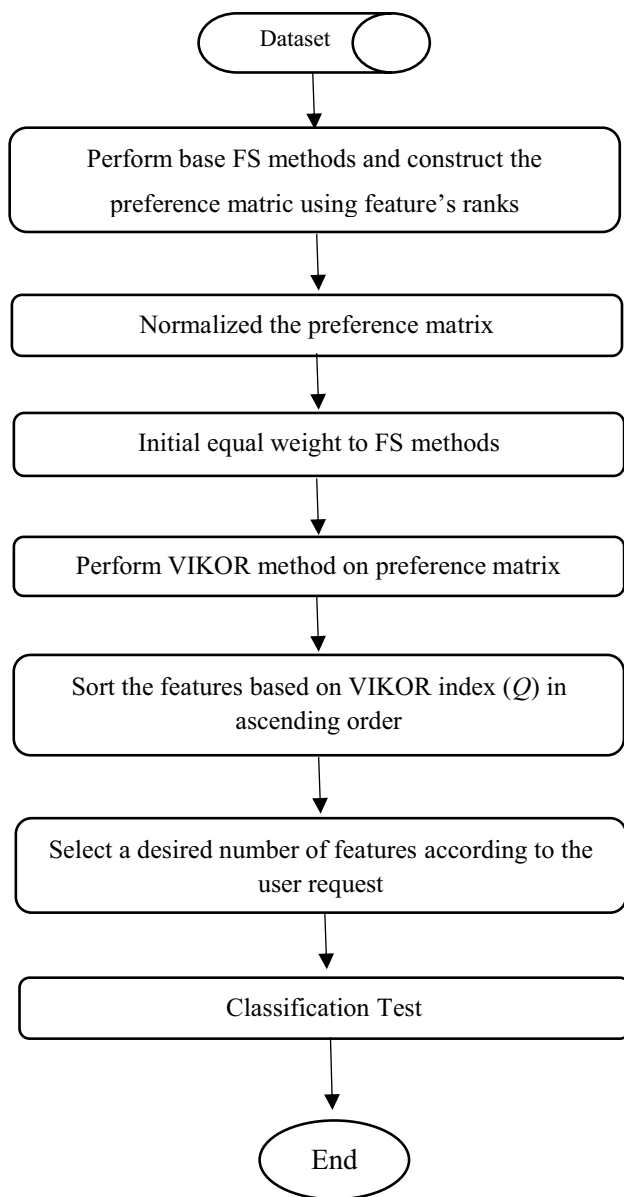


Fig. 2 Graphical abstract of EFS-MCDM

Q_i will tend to majority agreement and $v < 0.5$ then it will tend to majority negative attitude. Usually, $v = 0.5$, because it will balance the orientation towards the majority agreement and majority negative attitude. Then, rank the alternatives based on Q_i in ascending order. The smallest the value of Q_i represents the best alternative. Because it provides a maximum group utility of the “majority” and a minimum individual regret of the “opponent”. In several studies, the alternatives are sorted based on S , R , and Q , but we just used Q in this article.

4 Proposed method

In this section, we discuss our proposed algorithm in detail. This algorithm is constructed based on a heterogeneous strategy. In Fig. 2, a graphical abstract of the proposed method is presented. In this method, we used a rank aggregation procedure and model this fusion to an MCDM problem. The VIKOR method is used to rank the features based on the judgment of multiple FS algorithms. Algorithm 2 shows the step-by-step procedure of the proposed method.

Now we describe the steps of Algorithm 2 as our proposed method. We define an empty vector (FW) as a feature ranking vector for adding features in the first step. In step 2, the FS methods apply to data to evaluate the features as our criteria in the MCDM process. Each FS method obtains a rank vector representing the judgment of that method in the feature evaluation process. The first element of each vector is the index of the highest rank feature. The preference matrix is constructed based on these vectors as follows:

$$P = \begin{bmatrix} r_{11} & r_{21} & r_{31} & \cdots & r_{T1} \\ r_{12} & r_{22} & r_{32} & \cdots & r_{T2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{1M} & r_{2M} & r_{3M} & \cdots & r_{TM} \end{bmatrix}, \quad (7)$$

where each column is representing the vectors obtained from the last steps, and the rows are referring to the features. Also, r_{11} refers to the index of highest rank feature assigned by the first FS method and r_{TM} indicates the lowest rank feature assigned by the T -th FS method.

Algorithm 2: The proposed ensemble of feature selection algorithms: A multi-criteria decision-making approach (EFS-MCDM)

Input: $N \times M$ feature data matrix F , $N \times 1$ label vector Y

Output: Feature ranking vector FW

1. $FW = \emptyset$
 2. Perform Multiple feature selection algorithms on F and Y . Then construct the preference matrix (P) based on feature ranks
 3. $[M, T] = \text{size}(P)$ # Calculate the dimensions of the preference matrix
 4. $D = \text{zeros}(M, T)$ # construct a zeros matrix based on the number of features (M) and number of FS methods (T)
 5. # Normalized the Preference-Matrix (P) based on feature ranks and construct the decision matrix
 6. **for** $i = 1:T$
 7. **for** $j = 1:M$
 8. $\text{temp} = P(j, i)$
 9. $X(\text{temp}, i) = (m+1)-j$
 10. **end**
 11. **end**
 12. Initialize the weight of all FS methods to $1/T$
 13. Calculate the VIKOR index (Q) of each feature using Algorithm 1;
 14. $FW = \text{Sort the features based on } Q \text{ in ascending order}$
-

Because we considered our optimization problem as a maximization approach, and the decision matrix is based on the ranks of features, so the decision matrix needs to be normalized. Therefore, in steps 5 to 11, we set a score for each feature according to that feature's ranking. The highest rank feature (feature whose value in matrix P is 1) gets the score equals to the M (the number of features). The second-highest rank feature (feature whose value in matrix P is 2) gets the score equals to the M minus one and so on until all features have assigned new scores. The normalized decision matrix (X) is constructed as follows:

$$X = \begin{bmatrix} x_{11} & x_{21} & x_{31} & \cdots & x_{T1} \\ x_{12} & x_{22} & x_{32} & \cdots & x_{T2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{1M} & x_{2M} & x_{3M} & \cdots & x_{TM} \end{bmatrix}. \quad (8)$$

Now we have our decision matrix for our MCDM feature selection process in which the rows of matrix X (features) are considered as alternatives and the columns (FS methods)

as criteria (experts). The MCDM functions need the weight of criteria, so we set an equal weight to each criterion ($1/T$). In step 16, we used the VIKOR method to rank the features based on matrix X . We used Algorithm 1 to calculate the VIKOR index (Q) for each feature.

The results of applying the VIKOR algorithm is obtained as follows:

$$Q_i = \{Q_1, Q_2, \dots, Q_M\}, \quad (9)$$

where i is the feature number.

Finally, at the last step of the algorithm, the features are Sorted based on their value in Q in ascending order and save in a vector called FW . The user can select the desired number of features of vector FW . In Fig. 3, we used a sample data set with 5 features, 3 instances, and 4 FS methods to show our proposed algorithm's steps with an example.

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

a Sample dataset

$$\begin{array}{llll} \text{FS 1} = & \text{FS 2} = & \text{FS 3} = & \text{FS 4} = \\ \begin{bmatrix} 0.5 \\ 0 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} & \begin{bmatrix} 1000000 \\ 1000000 \\ 1000000 \\ 0.1111 \\ 0.1111 \end{bmatrix} & \begin{bmatrix} 0.25 \\ 0 \\ 0.25 \\ 0.25 \\ 0.25 \end{bmatrix} & \begin{bmatrix} 0.9183 \\ 0 \\ 0.9183 \\ 0.2516 \\ 0.2516 \end{bmatrix} \end{array}$$

b Scores assign to features based on 4 FS methods

$$\begin{array}{ll} R1 = \{2,1,3,4,5\} & R2 = \{1,2,3,4,5\} \\ R3 = \{1,3,4,5,2\} & R4 = \{1,3,4,5,2\} \end{array}$$

c Ranking vectors based on 4 rankers

$$P = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 3 & 3 \\ 3 & 3 & 4 & 4 \\ 4 & 4 & 5 & 5 \\ 5 & 5 & 2 & 2 \end{bmatrix}$$

d Feature Ranking Matrix (P)

$$X = \begin{bmatrix} 4 & 5 & 5 & 5 \\ 5 & 4 & 1 & 1 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 3 & 3 \\ 1 & 1 & 2 & 2 \end{bmatrix}$$

e Normalized Decision Matrix (X)

$$\begin{array}{l} f_j^* = \{5,5,5,5\} \\ f_j^- = \{1,1,1,1\} \end{array}$$

f Best and worst values for each FS method

$$S = \begin{bmatrix} 0.0625 & 0 & 0 & 0 \\ 0 & 0.0625 & 0.2500 & 0.2500 \\ 0.1250 & 0.1250 & 0.0625 & 0.0625 \\ 0.1875 & 0.1875 & 0.1250 & 0.1250 \\ 0.2500 & 0.2500 & 0.1875 & 0.1875 \end{bmatrix}$$

g Matrix S

$$S_i = \begin{bmatrix} 0.0625 \\ 0.05625 \\ 0.3750 \\ 0.6250 \\ 0.8750 \end{bmatrix} \quad R_i = \begin{bmatrix} 0.0625 \\ 0.2500 \\ 0.1250 \\ 0.1875 \\ 0.2500 \end{bmatrix}$$

h Utility measure (S) and Regret measure (R)

$$Q = \begin{bmatrix} 0 \\ 0.8077 \\ 0.3590 \\ 0.6795 \\ 1 \end{bmatrix}$$

i VIKOR index (Q)

$$FW = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 5 \\ 2 \end{bmatrix}$$

j Feature ranking vector**Fig. 3** Example of the steps of the proposed method

Table 1 Description of the datasets

Dataset	Instances	Features	References
Arcene	200	10,001	[69]
DLBCL	77	5470	[70]
Jaffe	213	677	[71]
ORL	400	1025	[72]
Pomeroy	60	7129	[73]
Semeion	1593	257	[69]
Shipp	77	7130	[70]
Lymphoma	96	4027	[5]
ALLAML	72	7130	[5]
Lung	203	3313	[5]

5 Experimental studies

To measure the performance of our proposed, we compared our method with four base FS methods: Fisher-Score [66], MIC [66], LLCFS [67], and CFS [68]. We have also compared our method to seven ensemble FS method: E-Borda [49], E-WBorda [49], E-Plu[49], E-Max [49], E-Min [49], E-Mean [49] and E-Median [49].

5.1 Datasets

We used 10 real-world datasets with a various number of features obtained to measure the performance of the proposed method with comparing methods. Table 1 contains a description of the following datasets.

5.2 Performance evaluation criteria

To measure the performance of the proposed EFS-MCDM and competing methods, we used accuracy [60] and F-score [50] metrics. First, consider the following concepts:

- *True positive (TP)*: percentage of positive examples correctly classified.
- *False-positive (FP)*: percentage of negative examples incorrectly classified as positive.
- *True negative (TN)*: percentage of negative examples correctly classified.
- *False-negative (FN)*: percentage of positive examples incorrectly classified as negative.

The accuracy and F-score are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

$$F - score = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \quad (11)$$

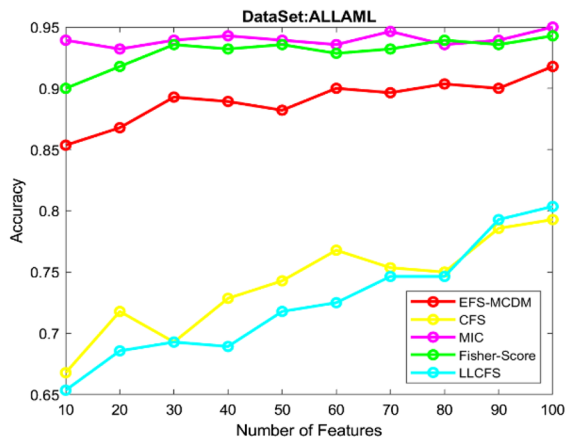
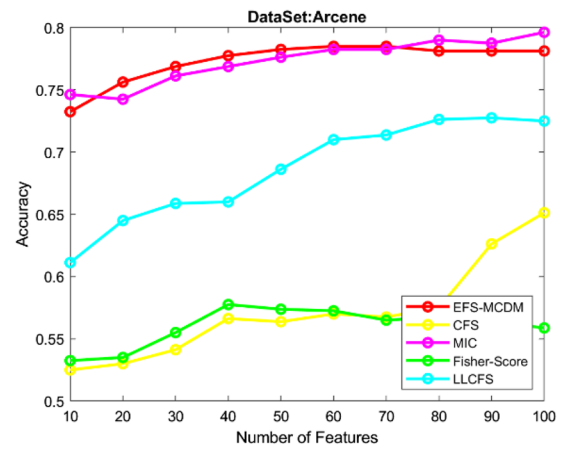
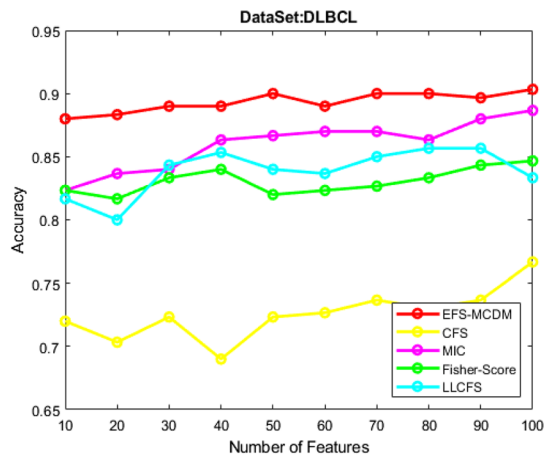
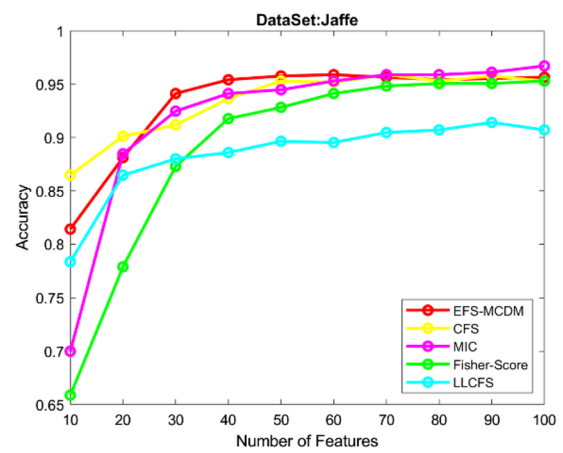
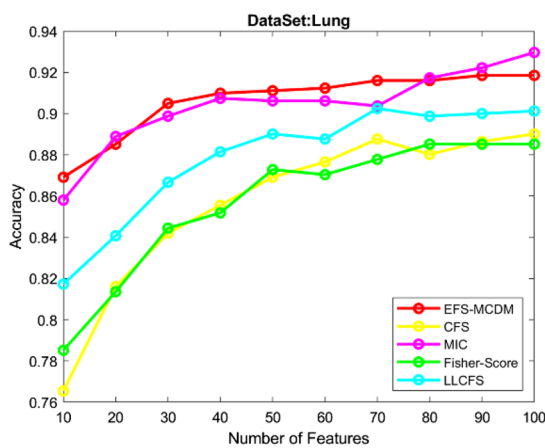
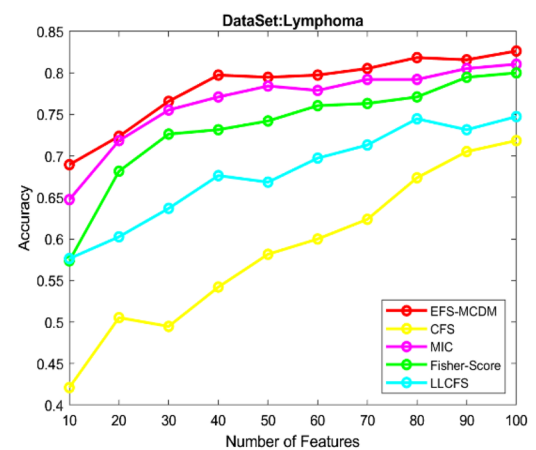
5.3 Results

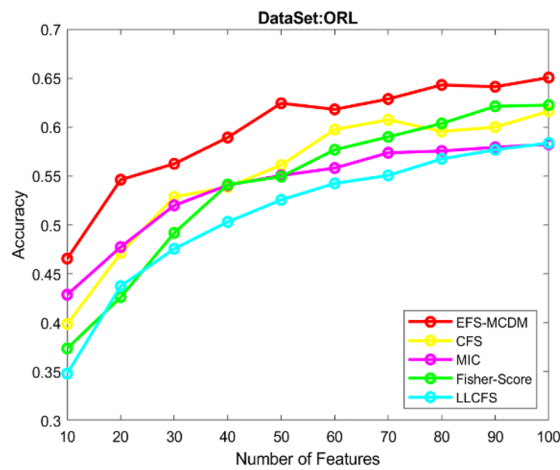
For all comparing methods, each parameter's value is set based on the recommendations by that corresponding paper. K-nearest neighbors (K-NN) classifier uses to compare the classification performance of the comparing algorithms and considered the number of neighbors equal to 5. We consider 60% of the instances randomly as training data for each test, while the remaining 40% regarded as test data. The user determines the number of selected features, and we change the size of the features subset from 10 to 100, which results in 100 different runs on each dataset. In our method, the user can determine the number of features. All the computations have been done on windows 8.1-64bit machine with Intel® Core (TM) i5-M460 and 4 GB Ram, using MATLAB® 9.4.0.813654 (R2018a).

5.3.1 Comparison between the proposed method and base FS methods

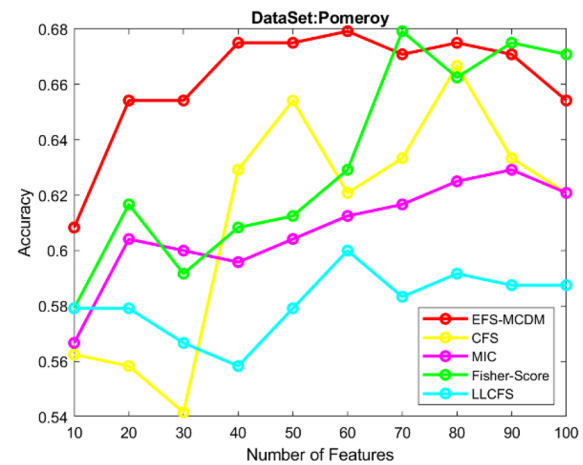
Figures 4 and 5 show the classification performance for accuracy and F-score criteria comparing based FS methods. In these figures, the horizontal axis shows the number of selected features. In these experiments, we have performed each algorithm 20 times on various feature subsets. The number of features in these subsets are then according to $m = \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. The vertical axis represents the result of performing the algorithm based on classification criteria. The algorithms are first evaluated by the top 10 features in their ranking system. Then, this procedure is continued for 20, 30, ..., 100 top features. These results are the average of 20 separate runs of each method.

We also compared the obtained results of EFS-MCDM and based FS methods statistically. To do this, we run the Friedman test [74] on the obtained results. The desired significance level for the post-hoc test is set to 0.05. If the result obtained by the Friedman test is less than the significance level, we perform a test for pairwise comparison of variables according to Conover [75]. The Friedman test results on EFS-MCDM against the other methods on 10 datasets are shown in the last row of Table 2. The values show the obtained *p*-values of each technique, comparing the EFS-MCDM method by the Friedman test. Also, the (+) sign indicates that our method statistically wins the comparing strategy, and respectively (=) sign and (−) sign show the tie and loss. The number of overall win/tie/loss of our method against others is shown in Table 3.

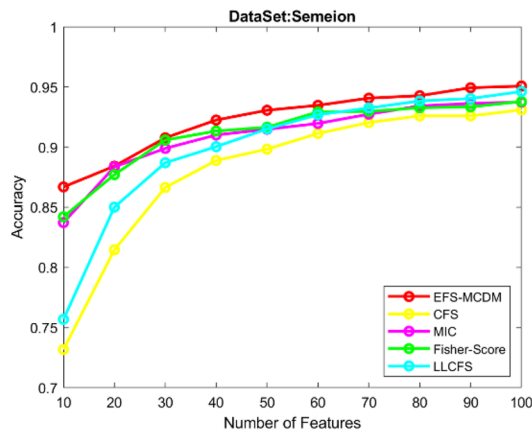
**a** ALLAML dataset**b** Arcene dataset**c** DLBCL dataset**d** Jaffe dataset**e** Lung dataset**f** Lymphoma dataset**Fig. 4** Comparison results based on accuracy metric



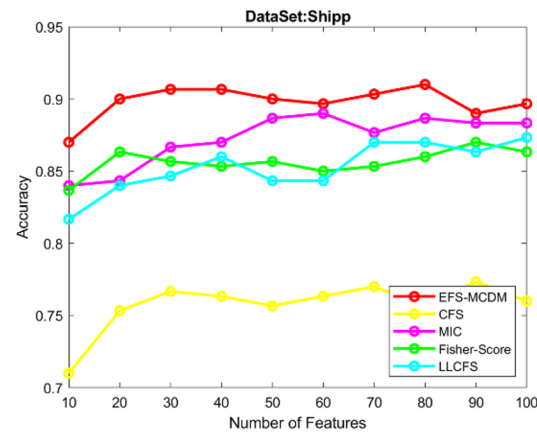
g ORL dataset



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i Semeion dataset



j Shipp dataset

Fig. 4 (continued)

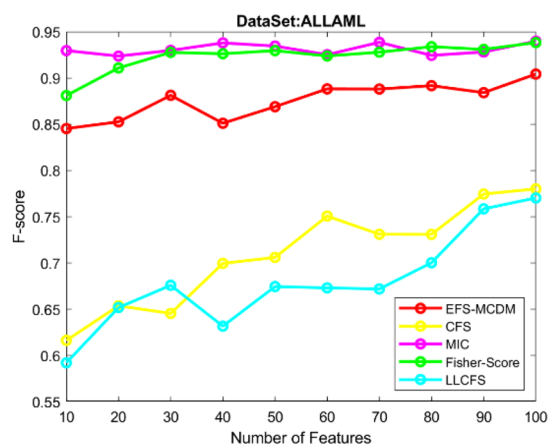
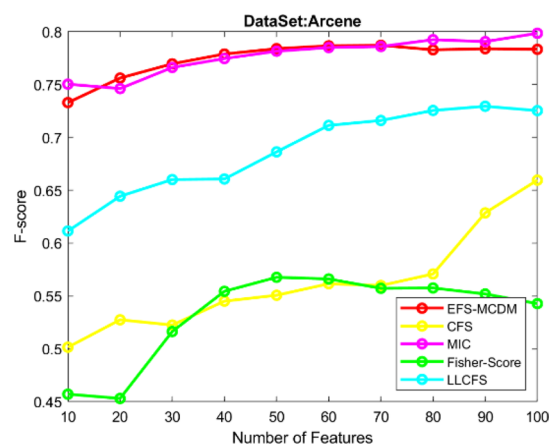
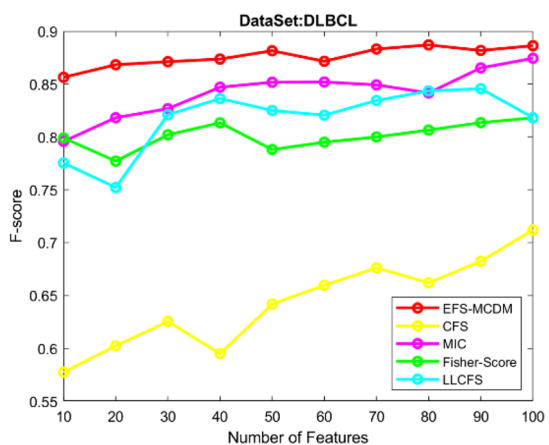
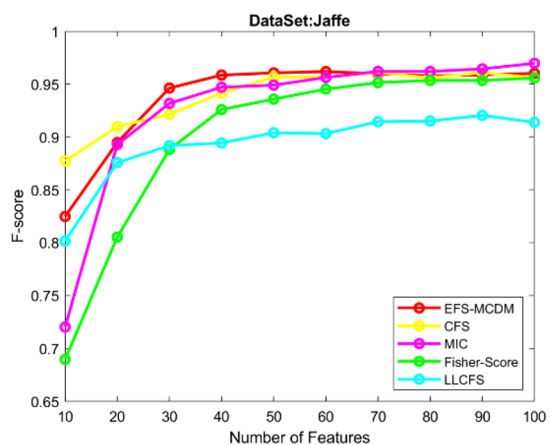
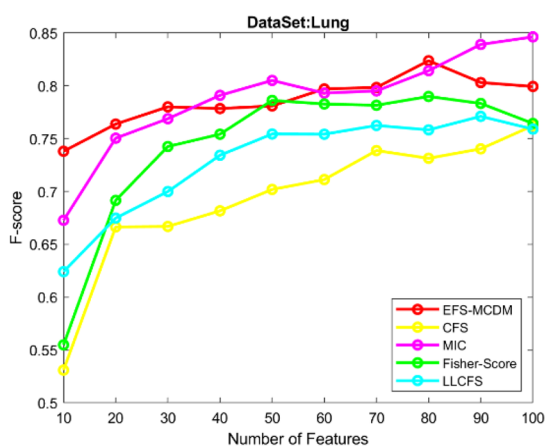
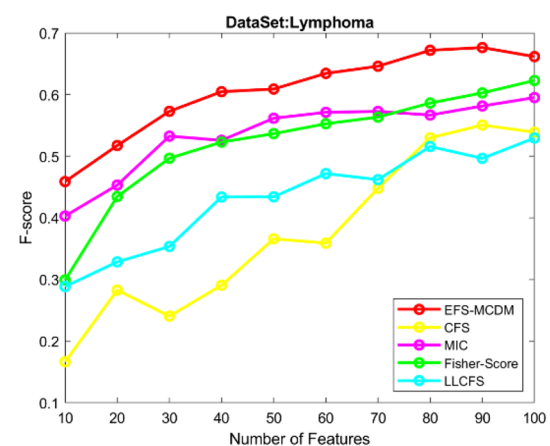
5.3.2 Comparison between the proposed method and ensemble FS methods

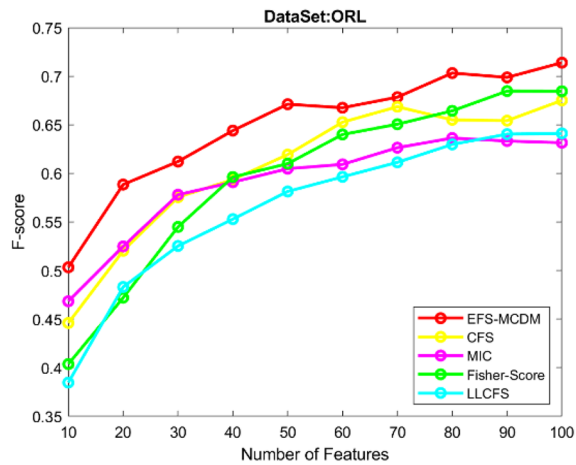
Tables 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 show the classification performance for accuracy and F-score criteria comparing ensemble FS methods. We also compared the obtained results of EFS-MCDM and other ensemble methods statistically. To do this, we run the Friedman test [74] on the obtained results. The desired significance level for the post-hoc test is set to 0.05. If the result obtained by the Friedman test is less than the significance level, we perform a test for pairwise comparison of variables according to Conover [75]. The Friedman test results on EFS-MCDM against the other methods on ten datasets are shown in the last row of Tables 4, 5, 6, 7, 8, 9, 10, 11, 12, 13. In the last row of tables,

the p-values of each method comparing the EFS-MCDM method obtained by the Friedman test are recorded. Also, (+) in the last row of each table indicates that our method statistically wins the comparing method and respectively (=) sign and (−) sign show the tie and loss. Finally, we showed the number of overall win/tie/loss of our method against others in Tables 14 and 15 represents the run-time of algorithms.

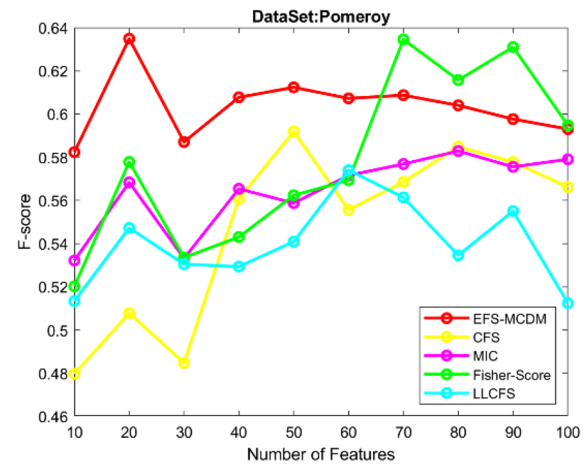
5.4 Discussions

In the suggested method, we treated the problem of ensemble feature selection as a Multi-Criteria Decision-Making problem for the first time. We considered the FS methods as our Decision Makers (DM) and tried to construct a ranking

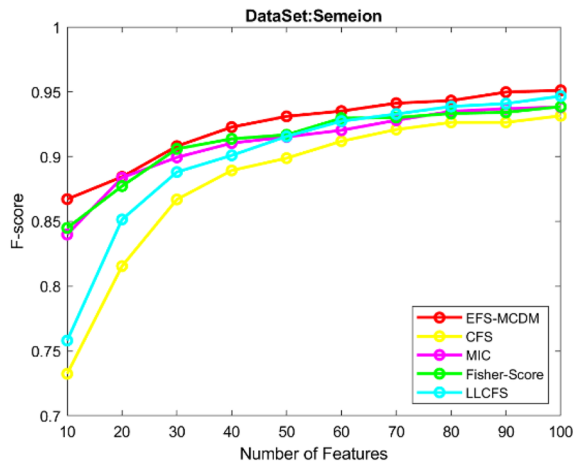
**a** ALLAML dataset**b** Arcene dataset**c** DLBCL dataset**d** Jaffe dataset**e** Lung dataset**f** Lymphoma dataset**Fig. 5** Comparison results based on F-score metric



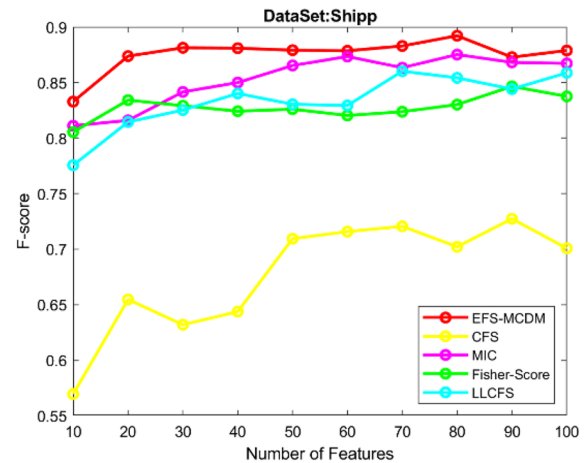
g ORL dataset



h Pomeroy dataset



i Semeion dataset



j Shipp dataset

Fig. 5 (continued)

system for features by the popular method called VIKOR. In the VIKOR method, the features are ranked based on each FS method's scores. Since the VIKOR method is an MCDM method, it evaluates the features found in their relationship with all FS methods. By the way, existing algorithms in ensemble feature selection still do not address all current challenges in ensemble FS. Among the weaknesses of existing algorithms, we can refer to low accuracy in high-dimensional datasets, long execution time, and tuned by specific feature selection methods. Thus, we have tried to present EFS-MCDM to improve the weakness of current algorithms. Based on the obtained results, we can see that our proposed model improves classification accuracy in high dimensional datasets like Shipp, Pomeroy, Arcene, and Lymphoma. Also, this algorithm does not have high computational complexity.

EFS-MCDM includes some simple and effective computations to overcome the long execution time in the ensemble process. On the other hand, this method is adaptive in the aggregation of various FS techniques, and some predefined techniques do not tune it.

The VIKOR algorithm requires $2 \times M \times T$ operations for calculating X^* and X' , $3 \times M$ operations for computing S , R , and Q , and 4 operations to calculate S^* , S^- , R^* , R^- . Also, $M \times T$ operations are needed for the normalization of the preference matrix. Thus, the time complexity of the proposed method will be:

$$2 \times M \times T + 3 \times M + 4.$$

Table 2 The obtained p-values by the Friedman test in terms of accuracy and F-Score against based FS methods

Dataset	EFS-MCDM VS CFS		EFS-MCDM VS LLCFS		EFS-MCDM VS Fisher-Score		EFS-MCDM VS MIC	
	Accuracy	F-score	Accuracy	F-score	Accuracy	F-score	Accuracy	F-score
Arcene	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	1(=)	0.729(=)
DLBCL	0.0004(+)	0.0005(+)	0.0004(+)	0.0005(+)	0.0004(+)	0.0005(+)	0.0004(+)	0.0005(+)
Jaffe	0.2987(=)	0.2987(=)	0.0018(+)	0.0018(+)	0.0005(+)	0.0005(+)	0.8597(=)	1(=)
ORL	0.0018(+)	0.0018(+)	0.0005(+)	0.0005(+)	0.0018(+)	0.0005(+)	0.0005(+)	0.0018(+)
Pomeroy	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0018(+)	0.0056(+)	0.0153(+)
Semeion	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0056(+)	0.0056(+)	0.0056(+)	0.0056(+)
Shipp	0.0005(+)	0.0005(+)	0.0004(+)	0.0005(+)	0.0005(+)	0.0018(+)	0.0004(+)	0.0005(+)
ALLAML	0.0004(−)	0.0005(−)	0.0004(+)	0.0005(+)	0.0004(+)	0.0056(+)	0.729(=)	0.729(=)
Lymphoma	0.0004(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0004(+)	0.0005(+)
Lung	0.0004(+)	0.0005(+)	0.0004(+)	0.0005(+)	0.0004(+)	0.0056(+)	0.729(=)	0.729(=)

The overall time complexity of the proposed method based on “Big Omicron” notation is $O(MT)$, where M is the number of features, and T is the number of FS methods.

The proposed method is scalable. To show that, we have done our experimental studies on datasets with various dimensions. For example, we used datasets with 257 features (Semeion) to 10,001 features (Arcene). This method performed well on all these datasets in a short time. In terms of being explainable, this model used a simple model with simple computations. MCDM is a process of choosing alternatives based on the judgments of several decision-makers. Thus, we assumed the ensemble feature selection as an MCDM problem where the FS methods are considered our decision-makers.

We classified the results of the proposed method compared to other methods into two groups. The first group is the results of comparing the proposed method against based FS methods. The results of this category are presented in Figs. 4 and 5. The second group is the results of comparing the proposed method against ensemble FS methods. These results are also presented in Tables 4, 5, 6, 7, 8, 9, 10, 11, 12, 13. Also, the overall win/tie/loss are presented in Table 14, and the averaged run-time of algorithms is recorded in Table 15. The results show that our method is superior to other methods in all evaluation criteria. According to the values in Table 15, the proposed method is so much faster than all ensemble methods. This matter is that our proposed

method does not use iterative methods for the ensemble process, and the VIKOR method is also a fast algorithm.

One of the issues that have not been reviewed in ensemble feature selection so far is to consider the different weights for the feature selection methods used in the aggregation process. This is one of our goals to improve the proposed method by providing a proper weighting method to improve the performance of the algorithm. For example, the strategies presented in these references [76, 77] can be tested for this purpose.

6 Conclusions and future works

In this work, we proposed a new feature selection algorithm for ensemble learning called, the EFS-MCDM which maps the features selection process to a Multi-Criteria Decision-Making problem. This method is based on a filter-based strategy and used a heterogeneous approach for ensemble learning. In this method, after we obtained the ranks of features based on multiple FS methods and construct a decision matrix, we deliver this data as our decision-making data to the VIKOR method to rank the features based on the evaluation of our experts. The results of different datasets show the optimality and efficiency of the proposed method.

One of our goals for improving the performance of the proposed method is to consider a dynamic weighting system. In this structure, the FS methods are not evaluated equally. We have referred to some potential techniques for this purpose in the last section. Also, we intend to use other approaches for ensemble FS, especially graph-based methods and extend our work to different types of feature selection like online feature selection.

Table 3 The win/tie/loss results of EFS-MCDM against the based FS methods based on Friedman test

Metric	EFS-MCDM method against			
	CFS	Fisher-Score	LLCFS	MIC
Accuracy	8/1/1	10/0/0	10/0/0	6/4/0
F-Score	8/1/1	10/0/0	10/0/0	6/4/0
Total	16/2/2	20/0/0	20/0/0	12/8/0

Table 4 Comparison results for DLBCL dataset

M/feat	Accuracy					F-Score										
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.8800	0.7433	0.8133	0.8433	0.7960	0.7967	0.7954	0.7970	0.7329	0.6651	0.3160	0.6980	0.6643	0.6623	0.6630	0.6593
20	0.8833	0.7833	0.8167	0.8700	0.8200	0.8200	0.8200	0.8210	0.7562	0.7104	0.3160	0.7369	0.7165	0.7145	0.7150	0.7225
30	0.8900	0.7733	0.8367	0.8967	0.8400	0.8400	0.8400	0.8408	0.7696	0.7155	0.3160	0.7730	0.7391	0.7371	0.7380	0.7431
40	0.8900	0.7633	0.8200	0.9200	0.8267	0.8267	0.8267	0.8273	0.7789	0.7151	0.3160	0.7970	0.7555	0.7535	0.7540	0.7605
50	0.9000	0.7533	0.8267	0.9333	0.8267	0.8267	0.8267	0.8275	0.7840	0.7465	0.3160	0.8005	0.7530	0.7510	0.7520	0.7600
60	0.8900	0.7667	0.8267	0.9300	0.8333	0.8333	0.8333	0.8343	0.7866	0.7385	0.3503	0.8220	0.7569	0.7549	0.7550	0.7509
70	0.9000	0.7900	0.8067	0.9267	0.8400	0.8400	0.8400	0.8412	0.7892	0.7419	0.3980	0.8010	0.7661	0.7641	0.7650	0.7541
80	0.9000	0.7733	0.8100	0.9067	0.8533	0.8533	0.8533	0.8541	0.7829	0.7351	0.5636	0.7901	0.7639	0.7619	0.7620	0.7559
90	0.8967	0.7833	0.8300	0.9133	0.8500	0.8500	0.8500	0.8511	0.7839	0.7506	0.6857	0.8021	0.76238	0.7612	0.7620	0.7568
100	0.9033	0.7967	0.8433	0.9200	0.8433	0.8433	0.8433	0.8439	0.7834	0.7465	0.7199	0.8012	0.7651	0.7631	0.7640	0.7581
Friedman Test		0.0004(+)	0.0004(+)	0.0399(-)	0.0004(+)	0.0004(+)	0.0004(+)	0.0004(+)		0.0005(+)	0.0005(+)	0.0399(-)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)

The best results are bold

Table 5 Comparison results for Arcene dataset

M/feat	Accuracy					F-Score										
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.7325	0.6637	0.4662	0.6900	0.6688	0.6488	0.6580	0.6588	0.7329	0.6681	0.3160	0.6908	0.6643	0.6623	0.6630	0.6633
20	0.7562	0.7087	0.4662	0.7300	0.7225	0.7025	0.7120	0.7125	0.7562	0.7104	0.3160	0.7298	0.7165	0.7145	0.7150	0.7155
30	0.7688	0.7137	0.4662	0.7488	0.7450	0.7250	0.7350	0.7350	0.7696	0.7155	0.3160	0.7430	0.7391	0.7371	0.7380	0.7381
40	0.7775	0.7113	0.4662	0.7550	0.7612	0.7412	0.7510	0.7512	0.7789	0.7151	0.3160	0.7529	0.7555	0.7535	0.7540	0.7545
50	0.7825	0.7438	0.4662	0.7537	0.7588	0.7388	0.7480	0.7488	0.7840	0.7465	0.3160	0.7551	0.7530	0.7511	0.7522	0.7520
60	0.7850	0.7375	0.4875	0.7762	0.7638	0.7738	0.7537	0.7538	0.7866	0.7385	0.3503	0.7759	0.7569	0.7549	0.7560	0.7559
70	0.7850	0.7388	0.5212	0.7787	0.7750	0.7550	0.7650	0.7650	0.7872	0.7419	0.3980	0.7769	0.7661	0.7651	0.7654	0.7651
80	0.7813	0.7337	0.6175	0.7775	0.7725	0.7525	0.7620	0.7625	0.7829	0.7351	0.5636	0.7790	0.7639	0.7619	0.7620	0.7629
90	0.7813	0.7512	0.6850	0.7913	0.7738	0.7538	0.7630	0.7638	0.7839	0.7506	0.6857	0.7920	0.7638	0.7618	0.7624	0.7628
100	0.7813	0.7475	0.7250	0.7788	0.7750	0.7550	0.7657	0.7650	0.7834	0.7465	0.7199	0.7760	0.7651	0.7631	0.7640	0.7641
Friedman Test		0.0005(+)	0.0004(+)	0.0056(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)		0.0005(+)	0.0004(+)	0.0056(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)

The best results are bold

Table 6 Comparison results for Jaffe dataset

M/feat	Accuracy				F-Score			
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.8141	0.8165	0.8859	0.8894	0.8788	0.8788	0.8788	0.8788
20	0.8812	0.8776	0.9224	0.9106	0.9059	0.9059	0.9059	0.9059
30	0.9412	0.9165	0.9329	0.9118	0.9165	0.9165	0.9165	0.9165
40	0.9541	0.9200	0.9341	0.9271	0.9188	0.9188	0.9188	0.9188
50	0.9567	0.9212	0.9412	0.9329	0.9224	0.9224	0.9224	0.9224
60	0.9588	0.9271	0.9365	0.9400	0.9306	0.9306	0.9306	0.9306
70	0.9565	0.9306	0.9435	0.9400	0.9306	0.9306	0.9306	0.9306
80	0.9541	0.9447	0.9459	0.9365	0.9376	0.9376	0.9376	0.9376
90	0.9533	0.9471	0.9506	0.9424	0.9424	0.9424	0.9424	0.9424
100	0.9565	0.9424	0.9518	0.9400	0.9424	0.9424	0.9424	0.9424
Friedman Test		0.0056(+)	0.0377(+)	0.0377(+)	0.0143(+)	0.0143(+)	0.0143(+)	0.0143(+)

The best results are bold

Table 7 Comparison results for ORL dataset

M/feat	Accuracy				F-Score			
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.4656	0.3969	0.4269	0.3688	0.3729	0.3709	0.3710	0.3719
20	0.5462	0.4963	0.5069	0.4644	0.4366	0.4346	0.4350	0.4356
30	0.5625	0.5312	0.5550	0.4794	0.4529	0.4509	0.4510	0.4519
40	0.5894	0.5569	0.5887	0.4994	0.4766	0.4746	0.4750	0.4756
50	0.6244	0.5819	0.6144	0.5219	0.5016	0.5000	0.5000	0.5006
60	0.6181	0.6012	0.6169	0.5475	0.5035	0.5015	0.5020	0.5025
70	0.6287	0.6131	0.6294	0.5575	0.5098	0.5078	0.5080	0.5088
80	0.6431	0.6169	0.6263	0.5619	0.5229	0.5209	0.5210	0.5219
90	0.6412	0.6269	0.6425	0.5575	0.5310	0.5300	0.5301	0.5300
100	0.6506	0.6319	0.6438	0.5631	0.5460	0.5440	0.5451	0.5450
Friedman Test		0.0018(+)	0.0833(=)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)

The best results are bold

Table 8 Comparison results for Pomeroy dataset

M/feat	Accuracy				F-Score											
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.6083	0.6333	0.6250	0.6083	0.6510	0.6500	0.6500	0.6500	0.5823	0.5719	0.5568	0.5472	0.5913	0.5903	0.5900	0.5903
20	0.6542	0.5875	0.6375	0.5750	0.6427	0.6407	0.6410	0.6417	0.6349	0.4971	0.5109	0.5253	0.5577	0.5557	0.5560	0.5567
30	0.6512	0.6000	0.5958	0.5917	0.6552	0.6532	0.6540	0.6542	0.5871	0.5098	0.4226	0.5284	0.6059	0.6039	0.6040	0.6049
40	0.6750	0.6083	0.6042	0.6000	0.6385	0.6365	0.6370	0.6375	0.6077	0.5423	0.4437	0.5277	0.6097	0.6077	0.6080	0.6087
50	0.6750	0.6167	0.6042	0.6292	0.6465	0.6445	0.6450	0.6455	0.6124	0.5481	0.4160	0.5821	0.6153	0.6133	0.6140	0.6143
60	0.6792	0.5917	0.6125	0.6292	0.6385	0.6365	0.6370	0.6375	0.6072	0.5231	0.4150	0.5829	0.6032	0.6012	0.6020	0.6022
70	0.6708	0.5750	0.6167	0.6333	0.6302	0.6282	0.6290	0.6292	0.6087	0.5086	0.4403	0.5771	0.5560	0.5540	0.5550	0.5550
80	0.6750	0.6000	0.6208	0.6208	0.6177	0.6157	0.6160	0.6167	0.6040	0.5584	0.4345	0.5724	0.5787	0.5767	0.5770	0.5777
90	0.6708	0.5958	0.6042	0.6292	0.6135	0.6115	0.6120	0.6125	0.5976	0.5296	0.4241	0.5797	0.5850	0.5830	0.5840	0.5840
100	0.6542	0.5917	0.6167	0.6250	0.6135	0.6115	0.6120	0.6125	0.5930	0.5245	0.4571	0.5752	0.5307	0.5307	0.5310	0.5317
Friedman Test		0.0018(+)	0.0056(+)	0.0005(+)	0.0087(+)	0.0087(+)	0.0087(+)	0.0087(+)		0.0005(+)	0.0005(+)	0.0005(+)	0.1659(=)	0.1659(=)	0.1659(=)	0.1659(=)

The best results are bold

Table 9 Comparison results for Semeion dataset

M/feat	Accuracy				F-Score											
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.8669	0.6958	0.7391	0.8378	0.5052	0.5032	0.5040	0.5042	0.5823	0.5719	0.5568	0.5472	0.5903	0.5903	0.5903	0.5903
20	0.8841	0.7783	0.8702	0.8721	0.5510	0.5500	0.5500	0.5500	0.6349	0.4971	0.5109	0.5253	0.5567	0.5567	0.5567	0.5567
30	0.9077	0.8251	0.9003	0.8954	0.5802	0.5782	0.5790	0.5792	0.5871	0.5098	0.4226	0.5284	0.6049	0.6049	0.6049	0.6049
40	0.9224	0.8440	0.9100	0.9096	0.5585	0.5565	0.5570	0.5575	0.6077	0.5423	0.4437	0.5277	0.6087	0.6087	0.6087	0.6087
50	0.9306	0.8724	0.9174	0.9220	0.6052	0.6052	0.6040	0.6042	0.6124	0.5481	0.4160	0.5821	0.6143	0.6143	0.6143	0.6143
60	0.9347	0.8793	0.9223	0.9331	0.6093	0.6073	0.6080	0.6083	0.6072	0.5231	0.4150	0.5829	0.6022	0.6022	0.6022	0.6022
70	0.9407	0.8895	0.9254	0.9425	0.6093	0.6073	0.6080	0.6083	0.6087	0.5086	0.4403	0.5771	0.5550	0.5550	0.5550	0.5550
80	0.9427	0.8981	0.9287	0.9440	0.6218	0.6198	0.6200	0.6208	0.6040	0.5584	0.4345	0.5724	0.5777	0.5777	0.5777	0.5777
90	0.9493	0.9064	0.9279	0.9469	0.6218	0.6208	0.6200	0.6208	0.5976	0.5296	0.4241	0.5797	0.5840	0.5840	0.5840	0.5840
100	0.9507	0.9097	0.8256	0.9493	0.6135	0.6115	0.6120	0.6125	0.5930	0.5245	0.4571	0.5752	0.5317	0.5317	0.5317	0.5317
Friedman Test		0.0056(+)	0.0056(+)	0.2207(=)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)		0.0005(+)	0.0056(+)	0.0005(+)	0.0377(+)	0.0377(+)	0.0377(+)	0.0377(+)

The best results are bold

Table 10 Comparison results for Shipp dataset

M/feat	Accuracy					F-Score										
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.8700	0.7133	0.7167	0.8233	0.7943	0.7923	0.7930	0.7933	0.8329	0.5854	0.6056	0.7905	0.7317	0.7317	0.7300	0.7307
20	0.9000	0.7500	0.7300	0.8733	0.8077	0.8057	0.8060	0.8067	0.8738	0.6289	0.6007	0.8451	0.7492	0.7472	0.7480	0.7482
30	0.9067	0.7500	0.7667	0.8533	0.8210	0.8210	0.8200	0.8200	0.8813	0.6531	0.6646	0.8401	0.7670	0.7650	0.7660	0.7660
40	0.9067	0.7667	0.7833	0.8567	0.8377	0.8357	0.8360	0.8367	0.8809	0.6829	0.6871	0.8398	0.7840	0.7820	0.7830	0.7830
50	0.9000	0.7600	0.7967	0.8633	0.8210	0.8210	0.8200	0.8200	0.8792	0.6849	0.7269	0.8325	0.7572	0.7552	0.7560	0.7562
60	0.8967	0.7833	0.7933	0.8600	0.8210	0.8220	0.8200	0.8200	0.8786	0.7341	0.7246	0.8520	0.7420	0.7380	0.7400	0.7400
70	0.9033	0.7767	0.7867	0.8767	0.8410	0.8420	0.8400	0.8400	0.8829	0.7298	0.6988	0.8520	0.7744	0.7714	0.7730	0.7734
80	0.9100	0.7867	0.7900	0.8883	0.8443	0.8423	0.8430	0.8433	0.8722	0.9408	0.7062	0.8520	0.7791	0.7751	0.7780	0.7781
90	0.8900	0.7700	0.7633	0.8700	0.8377	0.8357	0.8360	0.8367	0.8729	0.7135	0.6709	0.8730	0.7711	0.7711	0.7700	0.7701
100	0.8967	0.7733	0.7533	0.8800	0.8510	0.8510	0.8500	0.8500	0.8789	0.7089	0.6675	0.8603	0.7995	0.7975	0.7980	0.7985
Friedman Test		0.0005(+)	0.0005(+)	0.0016(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)		0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)

The best results are bold

Table 11 Comparison results for ALLAML dataset

M/feat	Accuracy					F-Score										
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.8679	0.5150	0.5050	0.4550	0.5060	0.5040	0.5052	0.5050	0.8578	0.5381	0.5288	0.4480	0.5381	0.5361	0.5370	0.5371
20	0.8929	0.5850	0.5200	0.6250	0.6260	0.6240	0.6251	0.6250	0.8888	0.6089	0.5303	0.6790	0.6223	0.6203	0.6210	0.6213
30	0.8750	0.5750	0.5450	0.6150	0.6160	0.6140	0.6153	0.6150	0.8703	0.5969	0.5459	0.6031	0.6469	0.6469	0.6450	0.6459
40	0.8786	0.5800	0.5500	0.6600	0.6610	0.6580	0.6602	0.6600	0.8802	0.5944	0.5493	0.6521	0.6979	0.6959	0.6960	0.6969
50	0.8821	0.6250	0.5650	0.6500	0.6560	0.6530	0.6551	0.6550	0.8836	0.6236	0.5443	0.6470	0.6935	0.6915	0.6920	0.6925
60	0.8964	0.6300	0.5800	0.6800	0.6810	0.6780	0.6802	0.6800	0.8972	0.6327	0.5695	0.6691	0.7339	0.7319	0.7320	0.7329
70	0.8786	0.6350	0.5850	0.6750	0.6760	0.6730	0.6752	0.6750	0.8850	0.6440	0.5829	0.6693	0.7315	0.7315	0.7300	0.7305
80	0.8786	0.6550	0.5950	0.6550	0.6560	0.6520	0.6555	0.6550	0.8877	0.6668	0.5959	0.6532	0.6962	0.6942	0.6950	0.6952
90	0.8750	0.6250	0.5850	0.6700	0.6710	0.6670	0.6702	0.6700	0.8770	0.6354	0.5938	0.6641	0.6693	0.6673	0.6680	0.6683
100	0.8857	0.6350	0.6200	0.6650	0.6660	0.6620	0.6651	0.6650	0.8857	0.6411	0.6345	0.6573	0.6899	0.6877	0.6890	0.6897
Friedman Test		0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)		0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)

The best results are bold

Table 12 Comparison results for Lymphoma dataset

M/feat	Accuracy					F-Score										
	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	EFS-MCDM	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean
10	0.6298	0.5150	0.5050	0.4550	0.5060	0.5040	0.5057	0.5050	0.4236	0.5381	0.5288	0.4236	0.5371	0.5361	0.5370	0.5371
20	0.8289	0.5850	0.5200	0.4800	0.6260	0.6240	0.6258	0.6250	0.4950	0.6089	0.5303	0.4950	0.6223	0.6203	0.6210	0.6213
30	0.7447	0.5750	0.5450	0.4900	0.6160	0.6120	0.6157	0.6150	0.5181	0.5969	0.5459	0.5181	0.6460	0.6459	0.6450	0.6459
40	0.7632	0.5800	0.5500	0.5400	0.6620	0.6610	0.6603	0.6600	0.5477	0.5944	0.5493	0.5477	0.6979	0.6959	0.6960	0.6969
50	0.7763	0.6250	0.5650	0.5400	0.6570	0.6530	0.6552	0.6550	0.5553	0.6236	0.5443	0.5553	0.6935	0.6915	0.6920	0.6925
60	0.8026	0.6300	0.5800	0.5400	0.6820	0.6810	0.6801	0.6800	0.5971	0.6327	0.5695	0.5791	0.7339	0.7319	0.7320	0.7329
70	0.8158	0.6350	0.5850	0.5700	0.6760	0.6770	0.6755	0.6750	0.5984	0.6440	0.5829	0.5984	0.7305	0.7315	0.7300	0.7305
80	0.8158	0.6550	0.5950	0.5950	0.6560	0.6540	0.6554	0.6550	0.6037	0.6668	0.5959	0.6037	0.6962	0.6942	0.6950	0.6952
90	0.8079	0.6250	0.5850	0.5700	0.6720	0.6710	0.6704	0.6700	0.5933	0.6345	0.5938	0.5933	0.6693	0.6663	0.6680	0.6683
100	0.8026	0.6350	0.6200	0.6150	0.6660	0.6660	0.6655	0.6650	0.5906	0.6411	0.6345	0.5906	0.6899	0.6877	0.6890	0.6897
Friedman Test		0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)	0.0005(+)		0.0005(-)	0.4884(=)	0.0005(+)	0.0005(-)	0.0005(-)	0.0005(-)	0.0005(-)

The best results are bold

Table 13 Comparison results for lung dataset

M/feat	Accuracy					F-Score										
	EFS-MCDM					EFS-MCDM					EFS-MCDM					
	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean	E-Bords	E-WBords	E-Plu	E-Max	E-Min	E-Median	E-Mean		
10	0.8691	0.8147	0.8407	0.8491	0.8471	0.8480	0.8481	0.7381	0.5669	0.7035	0.7381	0.6741	0.6731	0.6730	0.6731	
20	0.8852	0.8420	0.8605	0.8751	0.8731	0.8740	0.8741	0.7638	0.6277	0.7863	0.7638	0.7762	0.7742	0.7750	0.7752	
30	0.9049	0.8667	0.8580	0.8936	0.8916	0.8920	0.8926	0.7801	0.6935	0.7593	0.7801	0.8057	0.8037	0.8040	0.8047	
40	0.9099	0.8753	0.8790	0.9084	0.9064	0.9070	0.9074	0.7785	0.6987	0.7973	0.7785	0.8401	0.8381	0.8390	0.8391	
50	0.9111	0.8926	0.8778	0.8815	0.9109	0.9079	0.9099	0.7809	0.7286	0.8070	0.7809	0.8546	0.8526	0.8530	0.8536	
60	0.9123	0.8914	0.8926	0.8988	0.9146	0.9116	0.9130	0.7971	0.7135	0.8435	0.7981	0.8608	0.8588	0.8590	0.8598	
70	0.9160	0.8963	0.9086	0.9000	0.9195	0.9165	0.9180	0.7984	0.7334	0.8617	0.7984	0.8670	0.8650	0.8650	0.8660	
80	0.9160	0.8988	0.9160	0.9062	0.9279	0.9249	0.9250	0.8237	0.7381	0.8634	0.8237	0.8755	0.8735	0.8740	0.8745	
90	0.9185	0.9074	0.9160	0.9074	0.9245	0.9225	0.9230	0.8031	0.7445	0.8620	0.8031	0.8774	0.8754	0.8760	0.8764	
100	0.9185	0.9074	0.9247	0.9074	0.9245	0.9215	0.9230	0.7994	0.7459	0.8731	0.7994	0.8761	0.8741	0.8750	0.8751	
Friedman Test	0.0004(+)	0.0047(+)	0.0013(+)	0.4749(=)	0.4749(=)	0.4749(=)	0.4749(=)	0.0005(+)	0.0377(+)	0.0005(+)	0.0056(−)	0.0056(−)	0.0056(−)	0.0056(−)	0.0056(−)	

The best results are bold

Table 14 The win/tie/loss results of EFS-MCDM against the ensemble FS methods based on Friedman test

Metric	EFS-MCDM method against						
	E-Borda	E-WBorda	E-Plu	E-Min	E-Max	E-Median	E-Mean
Accuracy	10/0/0	9/1/0	8/1/1	9/1/0	9/1/0	9/1/0	9/1/0
F-Score	9/0/1	8/2/0	8/1/1	7/1/2	7/1/2	7/1/2	7/1/2
Total	19/0/1	17/3/0	16/2/1	16/2/2	16/2/2	16/2/2	16/2/2

Table 15 The Average Runtime of EFS-MCDM against the ensemble FS methods

Datasets	Methods							
	EFS-MCDM	E-Borda	E-WBorda	E-Plu	E-Min	E-Max	E-Median	E-Mean
Arcene	26.379	9906	9990	9860	9807	9810	9820	9818
DLBCL	4.056	1670	1791	1688	1680	1671	1674	1676
Jaffe	4.018	229	233	230	225	224	228	227
ORL	24.101	2008	2013	2010	2008	2007	2009	2010
Pomeroy	4.938	3059	3063	3057	3053	3050	3059	3060
Semeion	28.235	3763	3770	3760	3759	3757	3761	3762
Shipp	5.073	3250	3441	2775	2774	2770	2698	2778
ALLAML	4.845	2995	3410	2680	2770	2650	2590	2721
Lymphoma	3.012	1430	1410	1456	1478	1478	1475	1476
Lung	2.756	1281	1275	1245	1241	1243	1241	1239

The best results are bold

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